DAILY LESSON PLAN	School	**Taguig National High School**	Grade Level	**10**
	Teacher	**Ian B. Corocoto**	Learning Area	**SCIENCE (Earth Science)**
	Teaching Date and Time	**Day 4** **45 minutes**	Quarter	**Second Term (Week 9)**

I. OBJECTIVES		
A. Content Standards	Climate change and its impacts on the environment and people pose serious challenges which require solutions and action at local and global levels.	
B. Performance Standards	By the end of the term, learners gather information from secondary sources to describe rapid changes that are occurring in local and global climate patterns and propose solutions to address these changes at the local and global levels by drawing on awareness, responsible personal behavior to conserve materials and energy, and through the better societal management of the natural resources of the country.	
C. Most Essential Learning Competencies/ Objectives	At the end of the teaching-learning process, 100% of the students with at least 80% proficiency are expected to: 1. explain how El Niño and La Niña may impact the weather systems and daily life in the Philippines using local scenarios.	
II. CONTENT	**Global Interactions: ENSO Impact Scenario Activity – How Global Climatic Phenomena Affect Local Weather and Daily Life**	
III. LEARNING RESOURCES		
A. References	DepEd Grade 10 Science Budget of Work, Second Term, Week 9 – Global Interactions: "Describe how global climatic phenomena, such as the El Niño Southern Oscillation, may impact weather systems."; Suggested Activity: ENSO Impact Scenario Activity PAGASA (https://www.pagasa.dost.gov.ph) – ENSO Alert and Warning System; heat index categories NOAA Climate.gov – "El Niño & La Niña" (https://www.climate.gov/enso) SunStar Cebu – PAGASA definitions of dry condition, dry spell and drought (https://www.sunstar.com.ph/cebu/pagasa-cebu-under-dry-spellnot-drought) Materials: laptop and projector, three response cards per group (EL NIÑO / LA NIÑA / NEUTRAL), scenario cards, organizer sheets, manila paper, pentel pens	
IV. PROCEDURES	**Teacher's Activity**	**Student's Activity**
	Classroom Conditioning (Let the teacher count 1, 2, 3)	(As the teacher counts, the students will: 1- arrange the chairs, 2- pick up the pieces of paper, and 3- be ready to pray)
	Prayer Let us pray.	(The students will pray.)
	Greetings Good morning, class! Please be seated.	Good morning, Sir! It's nice to see you. Thank you.
	Checking of Attendance Who is absent for today?	The class secretary will report the attendance for today.
A. Reviewing previous lesson or presenting the new lesson (ELICIT) *4 minutes*	Let us play **"ENSO POP-UP."** I will read a clue. In your group, hold up the card **EL NIÑO, LA NIÑA,** or **NEUTRAL.** 1. "Trade winds are weaker than normal." 2. "Upwelling of cold water is stronger than normal." 3. "The Philippines tends to get below-normal rainfall." 4. "The warm pool sits in the western Pacific; winds are normal." 5. "Floods and landslides are more likely in many areas."	1. EL NIÑO 2. LA NIÑA 3. EL NIÑO 4. NEUTRAL 5. LA NIÑA (Groups raise their cards.)

	Excellent! Yesterday you were asked to bring a news story about drought, flood, or heat. Who would like to share the effect on people? (Two students share.)	(Two students share their news stories.)
B. Establishing a purpose for the lesson (ENGAGE) *2 minutes*	Today you will act as **PAGASA analysts and community advisers.** You will read real-life-style situations from different parts of the Philippines, decide what is happening in the Pacific, and advise the community. Our lesson objective is to: **explain how El Niño and La Niña may impact the weather systems and daily life in the Philippines using local scenarios.**	(The students read the objective.)
C. Presenting examples/ instances of the new lesson (EXPLORE) *10 minutes*	**ACTIVITY: "ENSO IMPACT SCENARIOS"** (Group of 5; 8 minutes; each group receives one scenario card, A–D) **Mechanics:** 1. Read your scenario card carefully. 2. Underline the **clues** (rainfall, temperature, water, crops, floods). 3. Complete the organizer: (a) ENSO phase and clues, (b) what changed in the local weather, (c) effects on daily life, (d) three actions (family, barangay, government). 4. Choose a reporter. Reports will be 1 minute each. (See the attached activity sheet for the scenarios.)	(The students receive their cards and begin.)
	(The teacher moves around and checks for understanding.) Guide questions: • What is the clue that tells you about rainfall? • What does the PAGASA statement in your card say? • Who is most affected: farmers, families, fishermen, students?	The rainfall clue is "far below normal for four months," so it is El Niño. Farmers and families are the most affected.
D. Discussing new concepts and practicing new skills #1 (EXPLAIN) *8 minutes*	Time is up. Let us hear from each group. (One reporter per scenario A–D presents.) **Scenario A – Water shortage:** What is the ENSO phase? How do you know?	El Niño. Rainfall was far below normal for four months, the dam level dropped, and PAGASA issued an El Niño advisory.
	How does the water shortage affect daily life, and what can families do?	Families queue for water, and cooking, laundry and hygiene are affected. Families can store water, fix leaks, and reuse water for cleaning.
	Scenario B – Rice farmers: What changed in the weather and what are the effects?	El Niño. Rainfall was low, canals dried, and yields dropped. Farmers can plant earlier or use drought-tolerant varieties, and the government can provide irrigation and assistance.
	Scenario C – Heat advisory: What is the phase? What does the heat index tell us?	El Niño. The heat index is in the danger range (42–51 °C), which can cause heat cramps, exhaustion and heatstroke. Students should drink water, stay in shade, and schools can adjust schedules.
	Scenario D – Flooding: What phase and which weather system changed?	La Niña. Rainfall was far above normal for weeks, so the river overflowed. Communities should clean drainage, prepare evacuation plans, and follow PAGASA warnings.
	Excellent work! Notice what these four scenarios show: a change in the **ocean and the winds** near the equator is **linked to weather thousands of kilometers away.** Scientists call these long-distance links **teleconnections.**	(The students listen and take notes.)

	The big idea: **global climatic phenomena such as ENSO shift the patterns of rising and sinking air and the location of rain, which can change local weather systems.** There are other global patterns too, such as monsoons and the Indian Ocean Dipole, but ENSO is the strongest influence on year-to-year climate in many places.	
E. Developing mastery (leads to Formative Assessment) (EXPLAIN) *2 minutes*	Can we say that El Niño or La Niña **caused** every dry or wet event we experience?	No, Sir. ENSO only changes the chances. Local weather and other factors also matter.
	Correct! ENSO makes some conditions more likely. That is why preparing early with PAGASA's advisories is important.	(The students nod.)
F. Finding practical application of concepts and skills in daily living (ELABORATE) *4 minutes*	**"30-SECOND ADVISORY"** With your group, write a 2–3 sentence **public advisory** for your scenario, as if you are announcing it on the radio or in a barangay Facebook page. Include: the ENSO phase, what weather to expect, and two actions. (Groups write for 2 minutes; two groups read aloud.)	Example (Scenario D): "Attention, residents! A La Niña Advisory is in effect. Heavy rains are expected in the coming weeks. Clean the canals, prepare your go-bag, and move to the evacuation center when told."
	How does a good public advisory help the community?	It tells people what is happening and what to do, so they can prepare early and stay safe.
G. Making generalizations and abstractions about the lesson (ELABORATE) *2 minutes*	Let us complete the thought: 1. ENSO is a natural cycle in which the _____ changes sea-surface temperature, winds, and rainfall. 2. During El Niño, the Philippines is usually _____, while during La Niña it is usually _____. 3. We should follow PAGASA advisories because _____.	1. Pacific Ocean. 2. Drier; wetter. 3. They help us prepare for dry spells, droughts, and floods.
H. Evaluating learning (EVALUATE) *8 minutes*	**MULTIPLE CHOICE:** Write the letter of the correct answer on one whole sheet of paper. 1. ENSO stands for: a) El Niño Southern Oscillation b) Eastern North Sea Oscillation c) Equatorial Northern Storm Origin d) Earth's Natural Seasonal Order 2. In which ENSO phase are the trade winds weaker than normal? a) La Niña b) El Niño c) Neutral only d) None 3. Which phase usually brings above-normal rainfall to many parts of the Philippines? a) El Niño b) La Niña c) Neither d) Both always 4. In normal conditions, where is the warm pool? a) Near South America b) In the western Pacific c) At the North Pole d) In the Indian Ocean only 5. Which is most likely during El Niño? a) Dry spell or drought b) More snow c) Stronger upwelling near Peru d) Flooding in all regions	1. A 2. B 3. B 4. B 5. A
	6. Three consecutive months of way-below-normal rainfall is described by PAGASA as:	6. C

	a) dry condition b) normal c) drought d) flood 7. A town has water rationing, cracked rice fields and heat index in the danger range. The most likely ENSO phase is: a) La Niña b) El Niño c) Neutral d) Cannot be determined 8. Why can ENSO, which starts in the tropical Pacific, affect weather in the Philippines? a) It shifts the location of rising air and rain, and the Philippines lies on the western edge near the warm pool. b) It moves the Philippines. c) It changes the speed of Earth's rotation. d) It only affects ships. 9. Barangay officials receive a La Niña Watch. Which is the best preparation? a) Store more water for a long drought. b) Clean drainage and canals and prepare evacuation plans. c) Ignore the advisory. d) Burn trash near the river. 10. A classmate says, "El Niño caused every dry day this year." Which response is best? a) Correct; El Niño controls all weather. b) Not necessarily; El Niño changes the chances of dry conditions, but local weather and other factors also matter. c) Wrong; El Niño has no effect on weather. d) Correct; La Niña does the same.	7. B 8. A 9. B 10. B
I. Additional activities for application or remediation **(EXTEND)** *1 minute*	**Assignment (Performance Task):** Make a **"Community ENSO Advisory"** on one-fourth cartolina or A4 paper (poster or infographic). Include: (1) the ENSO phase and a simple diagram, (2) the weather effects in the Philippines, and (3) at least three actions a family or barangay can take. This will be the bridge to our next topic, **Global and Local Sustainability.** The rubric is attached. Thank you, class! Goodbye.	(The students write the assignment in their notebooks.) Goodbye and thank you, Sir!
J. REFLECTION		
A. No. of learners who earned at least 80% in the evaluation		
B. No. of learners who require additional activities for remediation		
C. Did the remedial lessons work? No. of learners who have caught up with the lesson		
D. No. of learners who continue to require remediation		
E. Which of my teaching strategies worked well? Why did these work?		
F. What difficulties did I encounter which my principal or supervisor can help me solve?		

G. What innovation or localized materials did I use/discover which I wish to share with other teachers?

Prepared by:

IAN B. COROCOTO
Student Teacher

Checked:

[COOPERATING TEACHER]
Critic Teacher
Master Teacher 1

Noted:

[HEAD TEACHER]
Head Teacher III

Group: _____ Scenario: ____ Score: _____

"ENSO IMPACT SCENARIOS" – Scenario Cards

Card	Scenario (teacher-made, based on situations in the Philippines)
A – Water shortage	For four months, rainfall in a province has been far below normal. The dam supplying the water district has dropped to a very low level. The water district now supplies water only 6 hours a day. PAGASA has issued an El Niño advisory.
B – Rice farmers	In a rice-farming town, irrigation canals are almost dry and the soil in the fields is cracked. Many farmers delayed planting, and the harvest is expected to be much smaller than last year. PAGASA reports a dry spell in the area.
C – Heat advisory	For three weeks, the heat index in a city has reached the danger level (42–51 °C). Some schools have shifted classes to modular or online learning in the afternoon, and several students have been treated for heat exhaustion. PAGASA's El Niño advisory is in effect.
D – Flooding	For several weeks, rainfall in a river valley has been far above normal. The river has overflowed and flooded homes. Landslides blocked a road to the mountain barangays. Classes were suspended, and PAGASA has issued a La Niña advisory.

Group Organizer

Question	Our answer
(a) What ENSO phase is this? List 2–3 clues from the card.	
(b) What changed in the local weather (rainfall, temperature, storms)?	
(c) How does it affect daily life (water, food, health, school/work)?	
(d) Three actions: (1) family, (2) barangay, (3) government	

30-Second Advisory: _____

_____

Reference – PAGASA Heat Index Categories

Heat index	Classification	Possible effects
27–32 °C	Caution	Fatigue is possible with prolonged exposure and activity
33–41 °C	Extreme caution	Heat cramps and heat exhaustion are possible
42–51 °C	Danger	Heat cramps and heat exhaustion are likely; heatstroke is probable with continued exposure
52 °C and above	Extreme danger	Heatstroke is imminent

Rubric for the Scenario Activity (20 points)

Criteria	4	3	2	1
Correct ENSO phase with clues	Correct, 3 clues	Correct, 2 clues	Correct, 1 clue	Incorrect
Explanation of weather and daily-life effects	Complete and accurate	Mostly complete	Partly complete	Incorrect
Actions (family, barangay, government)	3 realistic actions	2 realistic actions	1 realistic action	None
30-second advisory	Clear, accurate and useful	Mostly clear	Unclear	Missing
Teamwork and reporting	Everyone participated; clear report	Most participated	Few participated	One member

Rubric for the Performance Task: "Community ENSO Advisory" (20 points)

Criteria	4	3	2	1
Scientific accuracy (ENSO phase, diagram)	Accurate and complete	Minor errors	Some errors	Many errors
Weather effects in the Philippines	3 or more correct effects	2 correct effects	1 correct effect	None
Actions for a family or barangay	3 or more realistic actions	2 realistic actions	1 realistic action	None
Creativity and layout	Original and attractive	Neat	Somewhat neat	Messy
Neatness, spelling and sources	Neat; correct; sources cited	Few errors	Several errors	Many errors

Answer Key – Day 4 Quiz

1. A 2. B 3. B 4. B 5. A 6. C 7. B 8. A 9. B 10. B